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Substrate-accommodating container

Abstract

The container main body includes a tubular wall portion having an opening circumferential portion provided at one end portion and the other end portion being closed, the opening circumferential portion having a container main body opening portion formed therein. The substrate storing space can store a substrate and communicates with the container main body opening portion. A central fixed member is provided to the wall portion of the container main body so as to be removably attached to the wall portion of the container main body. The central fixed member is configured to be locked and fixed to an alignment portion provided at a load port for conveying the substrate stored in the substrate storing space to a processing apparatus.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a substrate storing container for use in storing, keeping, conveying, transporting, and the like substrates composed of semiconductor wafers or the like.

BACKGROUND ART

(2) As a substrate storing container for storing and conveying substrates composed of semiconductor wafers, one has been known conventionally that includes a container main body and a lid body (see Patent Documents 1 and 2).

(3) The container main body includes a tubular wall portion in which a container main body opening portion is formed at one end portion, and in which the other end portion is closed. A substrate storing space is formed in the container main body. The substrate storing space is formed by being surrounded by the wall portion and can store a plurality of substrates. The lid body can be removably attached to the container main body opening portion and can close the container main body opening portion.

(4) A front retainer is provided at a portion of the lid body that faces a substrate storing space when the container main body opening portion is closed. When the container main body opening portion is closed by the lid body, the front retainer can support edge portions of the plurality of substrates. A back side substrate support portion is provided at the wall portion so as to form a pair with the front retainer. The back side substrate support portion can support edge portions of the plurality of substrates. When the container main body opening portion is closed by the lid body, the back side substrate support portion supports the plurality of substrates in cooperation with the front retainer, thereby retaining the plurality of substrates in a state in which the adjacent substrates are arranged in parallel to be spaced apart by a predetermined interval.

(5) The wall portion of the container main body includes a back wall, an upper wall, a lower wall, and a pair of side walls, and the container main body opening portion is formed by one end portion of the upper wall, one end portion of the lower wall, and one end portions of the side walls. The bottom portion of the container main body of the substrate storing container including the lower wall has abutting portions with which kinematic pins of a load port or the like for conveying the substrates stored in the substrate storing space to the processing apparatus abut. Further, the bottom portion has a central locked portion that is locked and fixed to a central container locking portion of a load port or the like. The central locked portion is positioned substantially equidistant from each of the abutting portions. The central locked portion is fixed to the central container locking portion of the load port or the like, so that the container main body is fixed to the load port or the like in a state of being supported by the kinematic pins of the load port or the like. Patent Document 1: PCT International Publication No. WO2007-034654 Patent Document 2: Japanese Unexamined Patent Application, Publication No. 2007-088064

DISCLOSURE OF THE INVENTION

Problems to be Solved by the Invention

(6) As described above, the central locked portion is locked to the central container locking portion of the load port or the like, so that the container main body is positioned and fixed to the load port or the like. Therefore, the central locked portion plays an important role in determining the center when the substrates are automatically transferred. Therefore, accuracy of the central locked portion needs to be improved, and the container needs to be fixed during operation of the load port. To fix the container, the container needs to be held to the lower face (reference face side) due to the structure of the central locked portion. The central locked portion has a hole shape and has a predetermined height in a Z direction (vertical direction).

(7) Therefore, when the container main body and the central locked portion are integrally molded,

the central locked portion becomes an undercut at the time of releasing the injection mold. Therefore, a complicated structure called a slide structure is required for the mold. As a result, since the operation of the mold becomes complicated, the operation speed needs to be reduced to prevent breakage of the mold. Thus, the molding cycle becomes long, resulting in lower production efficiency. Further, since the structure of the mold becomes complicated, it becomes difficult to form a cooling circuit for the mold for circulating cooling water in the mold to absorb the heat of the molten resin as an optimal circuit at an optimal position, and thus the cooling time is extended. (8) It is an object of the present invention to provide a substrate storing container that facilitates the formation of a cooling circuit for a mold for circulating cooling water in the mold as an optimal circuit at an optimal position by avoiding a complicated structure of the mold.

Means for Solving the Problems

(9) The present invention relates to a substrate storing container. The substrate storing container includes a container main body, a lid body, and a sealing member. The container main body includes a tubular wall portion having an opening circumferential portion provided at one end portion and the other end portion being closed, the opening circumferential portion having a container main body opening portion formed therein, and a substrate storing space formed by an inner face of the wall portion. The substrate storing space can store a substrate and communicates with the container main body opening portion. The lid body can be removably attached to the opening circumferential portion and can close the container main body opening portion by being in a positional relationship in which the lid body is surrounded by the opening circumferential portion. The sealing member is attached to the lid body and can abut with the lid body and the opening circumferential portion. The sealing member is configured to close the container main body opening portion in cooperation with the lid body by being interposed between the opening circumferential portion and the lid body so as to tightly abut with the opening circumferential portion and the lid body. A central fixed member is provided to the wall portion of the container main body so as to be removably attached to the wall portion of the container main body. The central fixed member is configured to be locked and fixed to an alignment portion provided at a load port for conveying the substrate stored in the substrate storing space to a processing apparatus.

(10) The central fixed member is preferably fixed to the wall portion of the container main body by a fitting structure.

(11) The central fixed member preferably includes a member front portion having an oval shape constituting a portion of the central fixed member on one end portion side of the container main body, and a member rear portion having a rectangular shape constituting a portion of the central fixed member on the other end portion side of the container main body.

(12) The member front portion is preferably engaged with and fixed to the container main body at one position, the member rear portion is preferably engaged with and fixed to the container main body at two positions, and the central fixed member is preferably fitted into the wall portion of the container main body.

(13) The central fixed member is preferably made of a polycarbonate resin. The central fixed member may be made of a cycloolefin polymer resin.

Effects of the Invention

(14) According to the present invention, it is possible to provide a substrate storing container that facilitates the formation of a cooling circuit for a mold for circulating cooling water in the mold as an optimal circuit at an optimal position by avoiding a complicated structure of the mold.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an exploded perspective view showing a substrate storing container 1 according to an

embodiment of the present invention;

(2) FIG. 2A is a right side view showing the substrate storing container **1** according to the embodiment of the present invention;

(3) FIG. 2B is a bottom view showing the substrate storing container **1** according to the embodiment of the present invention;

(4) FIG. 3 is an elevation view showing a container main body **2** of the substrate storing container **1** according to the embodiment of the present invention;

(5) FIG. 4 is an exploded perspective view showing the front portion of a central fixed member **6** provided to the container main body **2** of the substrate storing container **1** according to the embodiment of the present invention;

(6) FIG. 5 is an exploded perspective view showing the rear portion of the central fixed member **6** provided to the container main body **2** of the substrate storing container **1** according to the embodiment of the present invention;

(7) FIG. 6A is a perspective view showing the rear portion of the central fixed member **6** provided to the container main body **2** of the substrate storing container **1** according to the embodiment of the present invention; and

(8) FIG. 6B is a perspective view showing the front portion of the central fixed member **6** provided to the container main body **2** of the substrate storing container **1** according to the embodiment of the present invention.

PREFERRED MODE FOR CARRYING OUT THE INVENTION

(9) Hereinafter, a substrate storing container **1** according to an embodiment will be described with reference to the drawings. FIG. 1 is an exploded perspective view showing a state in which a substrate **W** is stored in the substrate storing container **1**. FIG. 2A is a right side view showing the substrate storing container **1**. FIG. 2B is a bottom view showing the substrate storing container **1**. FIG. 3 is an elevation view showing the container main body **2** of the substrate storing container **1**.

(10) Herein, for the purpose of illustration, the direction from a container main body **2** to a lid body **3** (described later) (direction from the upper right to the lower left in FIG. 1) is defined as a forward direction **D11** and the direction opposite to the direction **D11** is defined as a backward direction **D12**. These directions are collectively defined as a forward/backward direction **D1**. In addition, the direction from a lower wall **24** to an upper wall **23** (described later) (upper direction in FIG. 1) is defined as an upper direction **D21** and the direction opposite to the direction **D21** is defined as a lower direction **D22**. These directions are defined as an upper/lower direction **D2**. Moreover, the direction from a second side wall **26** to a first side wall **25** (described later) (direction from the lower right to the upper left in FIG. 1) is defined as a left direction **D31** and the direction opposite to the direction **D31** is defined as a right direction **D32**. These directions are defined as a left/right direction **D3**. The drawings show arrows indicating these directions.

(11) The substrate **W** (see FIG. 1) stored in the substrate storing container **1** is a disk-like silicon wafer, glass wafer, sapphire wafer, or the like, and is a thin member used for industrial use. The substrate **W** in the present embodiment is a silicon wafer having a diameter of 300 mm.

(12) As shown in FIG. 1, the substrate storing container **1** is used for storing the substrate **W** composed of a silicon wafer as described above and conveying it in a process in a factory as an in-process container or transporting the substrate by transport means such as land transport means, air transport means, or sea transport means as a shipping container. The substrate storing container **1** includes a container main body **2** and a lid body **3**. The container main body **2** includes a substrate support plate-like portion **5** and a back side substrate support portion **56** (see FIG. 3, etc.). The lid body **3** includes a front retainer (not shown).

(13) The container main body **2** includes a tubular wall portion **20** with a container main body opening portion **21** formed at one end portion and the other end portion closed. A substrate storing space **27** is formed in the container main body **2**. The substrate storing space **27** is surrounded by the wall portion **20**. The substrate support plate-like portion **5** is disposed at a portion of the wall

portion **20** that forms the substrate storing space **27**. As shown in FIG. **1**, a plurality of substrates **W** can be stored in the substrate storing space **27**.

(14) The substrate support plate-like portion **5** is provided at the wall portion **20** so as to form a pair in the substrate storing space **27**. When the container main body opening portion **21** is not closed by the lid body **3**, the substrate support plate-like portion **5** can support edge portions of the plurality of substrates **W** by abutting with the edge portions of the plurality of substrates **W** in a state in which the adjacent substrates **W** are arranged in parallel to be spaced apart by a predetermined interval. A back side substrate support portion **56** (see FIG. **3**, etc.) is provided at the back side of the substrate support plate-like portion **5**.

(15) The back side substrate support portion **56** is provided at the wall portion **20** so as to form a pair with the front retainer (not shown) described later in the substrate storing space **27**. When the container main body opening portion **21** is closed by the lid body **3**, the back side substrate support portion **56** can support the rear edge portions of the plurality of substrates **W** by abutting with the edge portions of the plurality of substrates **W**.

(16) The lid body **3** can be removably attached to an opening circumferential portion **28** (see FIG. **1**, etc.) forming the container main body opening portion **21** and can close the container main body opening portion **21**. The front retainer (not shown) is provided at a portion of the lid body **3** that faces the substrate storing space **27** when the container main body opening portion **21** is closed by the lid body **3**. The front retainer (not shown) is provided so as to form a pair with the back side substrate support portion **56** in the substrate storing space **27**.

(17) When the container main body opening portion **21** is closed by the lid body **3**, the front retainer (not shown) can support the front edge portions of the plurality of substrates **W** by abutting with the edge portions of the plurality of substrates **W**. When the container main body opening portion **21** is closed by the lid body **3**, the front retainer (not shown) supports the plurality of substrates **W** in cooperation with the back side substrate support portion **56**, thereby retaining the plurality of substrates **W** in a state in which the adjacent substrates **W** are arranged in parallel to be spaced apart by a predetermined interval.

(18) The substrate storing container **1** is made of a resin such as a plastic material or the like. If not specifically described, examples of the resin include thermoplastic resins such as polycarbonate, cycloolefin polymer, polyetherimide, polyetherketone, polybutyl terephthalate, polyetheretherketone, and liquid crystal polymer, and alloys thereof. In the case of imparting conductivity to these resins which are molding materials, conductive materials such as carbon fibers, carbon powder, carbon nanotubes, and conductive polymers are selectively added. It is also possible to add glass fibers, carbon fibers, or the like to increase the rigidity.

(19) Each portion is described in detail in the following. FIG. **4** is an exploded perspective view showing the front portion of a central fixed member **6** provided to the container main body **2** of the substrate storing container **1**. FIG. **5** is an exploded perspective view showing the rear portion of the central fixed member **6** provided to the container main body **2** of the substrate storing container **1**. FIG. **6A** is a perspective view showing the rear portion of the central fixed member **6** provided to the container main body **2** of the substrate storing container **1**. FIG. **6B** is a perspective view showing the front portion of the central fixed member **6** provided to the container main body **2** of the substrate storing container **1**.

(20) As shown in FIG. **1**, etc., the wall portion **20** of the container main body **2** includes a back wall **22**, an upper wall **23**, a lower wall **24**, a first side wall **25**, and a second side wall **26**. The back wall **22**, the upper wall **23**, the lower wall **24**, the first side wall **25**, and the second side wall **26** are made of the above-described material and are integrally molded.

(21) The first side wall **25** faces the second side wall **26**, and the upper wall **23** faces the lower wall **24**. The rear edge of the upper wall **23**, the rear edge of the lower wall **24**, the rear edge of the first side wall **25**, and the rear edge of the second side wall **26** are all connected to the back wall **22**. The front edge of the upper wall **23**, the front edge of the lower wall **24**, the front edge of the first side

wall **25**, and the front edge of the second side wall **26** have a positional relationship opposite the back wall **22**, and constitute the opening circumferential portion **28** that forms the container main body opening portion **21** in a substantially rectangular shape.

(22) The opening circumferential portion **28** is provided at one end portion of the container main body **2**, and the back wall **22** is located at the other end portion of the container main body **2**. The profile of the container main body **2** formed by the outer faces of the wall portion **20** is a box shape. The inner faces of the wall portion **20**, i.e., the inner face of the back wall **22**, the inner face of the upper wall **23**, the inner face of the lower wall **24**, the inner face of the first side wall **25**, and the inner face of the second side wall **26**, form the substrate storing space **27**, which is surrounded thereby. The container main body opening portion **21** formed by the opening circumferential portion **28** is in communication with the substrate storing space **27** surrounded by the wall portion **20** and formed inside the container main body **2**. The substrate storing space **27** can store a maximum of 25 substrates **W**.

(23) As shown in FIG. **1**, latch engagement concave portions **40A**, **40B**, **41A**, and **41B**, which are concave outwardly from the substrate storing space **27**, are formed at portions of the upper wall **23** and the lower wall **24** that are proximal to the opening circumferential portion **28**. A total of four latch engagement concave portions **40A**, **40B**, **41A**, and **41B** are respectively formed near both right and left end portions of the upper wall **23** and the lower wall **24**.

(24) As shown in FIG. **1**, a rib **235** is provided so as to be integrally formed with the upper wall **23** at the outer face of the upper wall **23**. The rib **235** increases the rigidity of the container main body **2**.

(25) A top flange **236** is fixed to the central portion of the upper wall **23**. The top flange **236** is a member corresponding to a portion of the substrate storing container **1** from which it is hung to be suspended when suspending the substrate storing container **1** by an automated material handling system (AMHS), a person guided vehicle (PGV), etc.

(26) At the central position of the lower wall **24**, there are provided the central fixed member **6** having a central through hole **601** formed therein, a front side fixing portion **245** for fixing the central fixed member **6**, a rear side fixing portion **246**, and a detection central convex portion **247** provided at a position between the front side fixing portion **245** and the rear side fixing portion **246**.

(27) The front side fixing portion **245** is provided at a position advanced from the central position of the lower wall **24** in the forward direction **D11** by approximately one third of the length between the central position of the lower wall **24** and the front edge of the lower wall **24**. The front side fixing portion **245** includes a rear wall **2451** extending in substantially the left/right direction **D3** and projecting in the lower direction **D22**, side walls **2452** extending in the forward direction from both left and right end portions of the rear wall **2451** and projecting in the lower direction **D22**, and a lower wall **2453** integrally formed with the lower edge of the rear wall **2451** and the lower edges of the side walls **2452** to be integrally connected. The front edges of the side walls **2452** and the front edge of the lower wall **2453** constitute an opening portion that opens toward the forward direction **11**. Through holes **2455** passing through in the forward/backward direction **D1** are formed at the central portion of the rear wall **2451** and both left and right sides thereof. A front portion engaged portion **613** (described later) is fitted into the central through hole **2455**, whereby the front portion engaged portion **613** is engaged with and fixed to the front side fixing portion **245** at this one position.

(28) The rear side fixing portion **246** is provided at a position advanced from the central position of the lower wall **24** in the backward direction **D12** by approximately one third of the length between the central position of the lower wall **24** and the rear edge of the lower wall **24**. The rear side fixing portion **246** includes a pair of side walls **2461** extending in substantially the forward/backward direction **D1** and projecting in the lower direction **D22**, and lower walls **2463** respectively extending from the rear portions of the pair of side walls **2461** so as to approach each other and be

integrally formed with the side walls **2461** to be integrally connected. The front end portion of the lower wall **2463** has an inclined face **2465** that rises in the upper direction **D21** toward the forward direction **D11**. At the lower side of the pair of lower walls **2463**, locked plate portions **6232** of a rear portion engaged portion **623** (described later) are respectively disposed, whereby the rear portion engaged portion **623** is engaged with and fixed to the rear fixing portion **246** at two positions, i.e., at both end portions of the rear portion engaged portion **623** in the left/right direction **D3**. The detection central convex portion **247** is constituted by rectangular wall portions extending in the left/right direction **D3** and projecting in the lower direction **D22**.

(29) The central fixed member **6** is locked and fixed to a boss constituting a container fixing portion provided at a load port (not shown) for conveying the substrates **W** stored in the substrate storing space **27** to a processing apparatus. The central fixed member **6** is composed of a component independent of the lower wall **24**, and is fixed to the lower wall **24** constituting the wall portion **20** of the container main body **2** by a fitting structure.

(30) The central fixed member **6** is made of a resin such as a polycarbonate resin, a cycloolefin polymer resin, a polybutylene terephthalate resin, a polyacetal resin, or a polyetheretherketone resin. In the present embodiment, a polycarbonate resin or a cycloolefin polymer resin is used.

(31) As shown in FIGS. **6A**, **6B**, etc., the central fixed member **6** includes a member front portion **61** and a member rear portion **62**. The member front portion **61** has an oval shape in bottom view, the member rear portion **62** has a rectangular shape in bottom view, and the member front portion **61** and the member rear portion **62** are integrally formed and integrally connected.

(32) A front portion peripheral wall portion **612** projecting in the upper direction **D21** is integrally formed with and integrally connected to a front portion bottom portion **611**, around the member front portion **61**. The front portion peripheral wall portion **612** is provided over the entire peripheral edge portion excluding the rear portion of the member front portion **61**. A central through hole **601** composed of a substantially rectangular through hole elongated in the forward/backward direction in plan view is formed at a position adjacent to the forward part of the front portion peripheral wall portion **612**. The boss constituting the container fixing portion of the load port (not shown) is inserted into the central through hole **601**, the central fixed member **6** is locked to the boss, and the container main body **2** is held and fixed to the reference surface side of the load port.

(33) At the front edge of the front portion peripheral wall portion **612**, a front portion engaged portion **613** projecting in the forward direction is integrally formed with and integrally connected to the front portion peripheral wall portion **612**. The front portion engaged portion **613** includes a plate-like base portion **6131** that is parallel to the forward/backward direction **D1** and the left-right direction **D3** and projects from the front edge of the front portion peripheral wall portion **612** in the forward direction **D11**, a pair of upper ribs **6132** that are integrally formed with the front edge of the front portion peripheral wall portion **612** and the upper face of the base portion **6131** and extend in the forward direction **D11**, and a pair of lower ribs **6133** that are integrally formed with the front edge of the front portion peripheral wall portion **612** and the lower face of the base portion **6131** and extend in the forward direction **D11**. The front edges of the upper rib **6132** and the lower rib **6133** are located further rearward than the front edge of the base portion **6131**. The portion of the front edge of the base portion **6131** where the upper rib **6132** and the lower rib **6133** are not provided is inserted into the through hole **2455** of the front side fixing portion **245**, and thereby the front portion engaged portion **613** is engaged with the front side fixing portion **245** to be locked and fixed.

(34) As shown in FIGS. **6A** and **6B**, a support wall portion **615** is provided on the upper face of the front portion bottom portion **611** of the member front portion **61**. The support wall portion **615** is composed of a substantially triangular wall portion extending in the forward/backward direction **D1** and projecting in the upper direction **D21**. The upper edge of the support wall portion **615** abuts with the lower wall **24** when the central fixed member **6** is fixed to the lower wall **24**.

(35) In the member front portion **61A**, a through hole **616** extending in the left/right direction is

formed rearward of the portion where the central through hole **601** is formed. The through hole **616** has a width and a length that are slightly longer than the width and the length of the detection central convex portion **247** in bottom view. When the central fixed member **6** is fixed to the lower wall **24**, the detection central convex portion **247** is inserted into the through hole **616**.

(36) At the rear edge of the member rear portion **62**, the rear portion engaged portion **623** projecting in the upper direction **D21** is integrally formed with and integrally connected to a rear portion bottom portion **621**. The rear portion engaged portion **623** includes a rectangular plate-like portion **6231** that is longer than the member rear portion **62** in the left/right direction **D3**. At the end portions of the upper end portion of the plate-like portion **6231**, the locked plate portions **6232**, which extend obliquely from the upper end edge of the plate-like portion **6231** in the backward direction **D12** and the downward direction **D22**, are integrally formed with the plate-like portion **6231** to be integrally connected. When the central fixed member **6** is fitted into the front side fixing portion **245** and the rear side fixing portion **246**, the locked plate portions **6232** slide on the inclined faces **2465**, pass over the front end edges of the inclined faces **2465**, enter the lower sides of the lower walls **2463**, and the rear portion engaged portion **623** is engaged with the rear side fixing portion **246** to be locked and fixed.

(37) The lower wall **24** constituting the bottom portion is provided with groove portions **7** serving as abutting portions. The groove portions **7** respectively abut with a plurality of kinematic pins. The kinematic pins are provided at the load port (not shown) for conveying the substrates **W** stored in the substrate storing space **27** to the processing apparatus, as a plurality of alignment portions, and align the container main body **2** with the load port. By abutting with the kinematic pins, the substrate storing container **1** is aligned in the forward/backward direction **D1**, the left/right direction **D3**, and the upper/lower direction **D2**.

(38) Three groove portions **7** are provided at three positions of the lower wall **24** at intervals forming a center angle of about 120° in bottom view along the outer face of the lower wall **24** around the front portion of the central through hole **601**. The groove portions **7** are formed such that the grooves of all of the groove portions **7** are directed toward the central through hole **601**.

(39) As shown in FIG. 1, etc., the lid body **3** has a substantially rectangular shape that substantially matches the shape of the opening circumferential portion **28** of the container main body **2**. The lid body **3** is removably attached to the opening circumferential portion **28** of the container main body **2**. By the lid body **3** being attached to the opening circumferential portion **28**, the lid body **3** can close the container main body opening portion **21**. An annular seal member **4** is attached to an inner face of the lid body **3** (a face on the back side of the lid body **3** shown in FIG. 1) that faces a face (a seal face **281**) of a step portion formed at a position immediately rearward (backward direction **D12**) of the opening circumferential portion **28** when the lid body **3** closes the container main body opening portion **21**. The seal member **4** may be made of various types of thermoplastic elastomers such as polyester type and polyolefin type that are elastically deformable, fluorine containing rubber, silicon rubber, or the like. The sealing member **4** is arranged so as to go around the circumferential edge portion of the lid body **3**.

(40) When the lid body **3** is attached to the opening circumferential portion **28**, the sealing member **4** is sandwiched between the sealing face **281** and the inner face of the lid body **3** to be elastically deformed, and thus the lid body **3** closes the container main body opening portion **21** in an airtight state. By the lid body **3** being removed from the opening circumferential portion **28**, it is possible to load or unload the substrates **W** relative to the substrate storing space **27** in the container main body **2**.

(41) A latching mechanism is provided at the lid body **3**. The latching mechanism is provided in the vicinity of both left and right end portions of the lid body **3**. As shown in FIG. 1, the latching mechanism includes two upper side latch portions **32A** and **32B** that can project from the upper side of the lid body **3** in the upper direction **D21** and two lower side latch portions (not shown) that can project from the lower side of the lid body **3** in the lower direction **D22**. The two upper side

latch portions **32A** and **32B** are arranged in the vicinity of both left and right ends of the upper side of the lid body **3**, and the two lower side latch portions are arranged in the vicinity of both left and right ends of the lower side of the lid body **3**.

(42) An operation portion **33** is provided at an outer face of the lid body **3**. By operating the operation portion **33** from the front side of the lid body **3**, it is possible to cause the upper side latch portions **32A** and **32B** and the lower side latch portions (not shown) to project from the upper side and the lower side of the lid body **3**, and it is also possible to cause the upper side latch portions **32A** and **32B** and the lower side latch portions (not shown) not to project from the upper side and the lower side of the lid body **3**. By the upper side latch portions **32A** and **32B** projecting from the upper side of the lid body **3** in the upper direction **D21** to engage with the latch engagement concave portions **40A** and **40B** of the container main body **2**, and the lower side latch portions (not shown) projecting from the lower side of the lid body **3** in the lower direction **D22** to engage with the latch engagement concave portions **41A** and **41B** of the container main body **2**, the lid body **3** is fixed to the opening circumferential portion **28** of the container main body **2**.

(43) On the inner side of the lid body **3** (opposite side to the outer face of the lid body **3** appearing in FIG. 1), a concave portion (not shown) which is concave outward of the substrate storing space **27** is formed. A front retainer (not shown) is fixed to the concave portion (not shown) and a portion outside the concave portion in the lid body **3**.

(44) The front retainer (not shown) includes front retainer substrate receiving portions (not shown). The front retainer substrate receiving portions (not shown) are disposed two by two so as to form a pair and at a predetermined interval in the left/right direction **D3**. The front retainer substrate receiving portions disposed two by two so as to form a pair as described above are provided in a state in which **25** pairs are arranged in parallel in the upper/lower direction **D2**. The front retainer substrate receiving portions sandwich and support the end edges of edge portions of the substrates **W** by the substrates **W** being stored in the substrate storing space **27** and the lid body **3** being closed.

(45) According to the substrate storing container **1** of the present embodiment having the above-described configuration, the following effects can be obtained.

(46) As described above, in the substrate storing container **1**, the central fixed member **6** is provided to the wall portion **20** of the container main body **2** so as to be removably attached to the wall portion **20** of the container main body **2**. The central fixed member **6** is configured to be locked and fixed to the boss constituting the container fixing portion provided at the load port for conveying the substrates **W** stored in the substrate storing space **27** to the processing apparatus.

(47) This configuration makes it possible to avoid causing an undercut when the injection mold for molding the container body **2** is released. This makes it possible to avoid providing a complicated structure called a slide structure in the mold, and to avoid complicating the operation of the mold. This eliminates the need to reduce the operation speed to prevent breakage of the mold, and makes it possible to avoid lengthening the molding cycle and lower production efficiency. Further, this makes it possible to avoid a complicated structure of the mold, and to form a cooling circuit for the mold for circulating cooling water in the mold to absorb the heat of the molten resin as an optimal circuit at an optimal position. This makes it possible to avoid extending the cooling time.

(48) The central fixed member **6** is fixed to the wall portion **20** of the container body **2** by a fitting structure. This makes it possible to easily realize a configuration in which the central fixed member **6** can be removably attached to the wall portion **20** of the container main body **2**.

(49) The central fixed member **6** includes the member front portion **61** having an oval shape constituting a portion of the central fixed member **6** on one end portion side of the container main body **2**, and the member rear portion **62** having a rectangular shape constituting a portion of the central fixed member **6** on the other end portion side of the container main body **2**. The member front portion **61** is engaged with and fixed to the container main body **2** at one position, the member rear portion **62** is engaged with and fixed to the container main body **2** at two positions,

and the central fixed member **6** is fitted into the wall portion **20** of the container main body **2**.
(50) This configuration enables the central fixed member **6** to be easily fitted into the lower wall **24** of the container body **2**. In addition, the above-described shape enables good fluidity of the resin and good moldability. Further, the above-described shape makes it possible to make the packaging bag less likely to break when the substrate storing container **1** is packaged and transported.
(51) The central fixed member **6** is made of a polycarbonate resin. The central fixed member **6** is made of a cycloolefin polymer resin. This configuration enables the central fixed member **6** to have a low water absorption rate, a low moisture absorption rate, and a low outgassing amount.
(52) The present invention is not limited to the above-described embodiment, and modifications thereto within the technical scope claimed in the claims are possible. For example, the shapes of the container main body and the lid body and the number and dimensions of the substrates **W** that can be stored in the container main body are not limited to the shapes of the container main body **2** and the lid body **3** and the number and dimensions of the substrates **W** that can be stored in the container main body **2** according to the present embodiment. For example, the configuration such as the shape of the central fixed member or the like is not limited to the configuration such as the shape of the central fixed member **6** according to the present embodiment.

EXPLANATION OF REFERENCE NUMERALS

(53) **1** substrate storing container **2** container main body **3** lid body **4** central fixed member **20** wall portion **21** container main body opening portion **22** back wall **23** upper wall **24** lower wall (bottom portion) **25** first side wall **26** second side wall **27** substrate storing space **61** member front portion **62** member rear portion **W** substrate

Claims

1. A substrate storing container, comprising: a container main body including a tubular wall portion having an opening circumferential portion provided at one end portion and the other end portion being closed, the opening circumferential portion having a container main body opening portion formed therein, and a substrate storing space formed by an inner face of the wall portion, the substrate storing space being capable of storing a substrate and communicating with the container main body opening portion; a lid body removably attached to the opening circumferential portion and closing the container main body opening portion by being in a positional relationship in which the lid body is surrounded by the opening circumferential portion; and a sealing member attached to the lid body and abutting with the lid body and the opening circumferential portion, the sealing member being configured to close the container main body opening portion in cooperation with the lid body by being interposed between the opening circumferential portion and the lid body so as to tightly abut with the opening circumferential portion and the lid body, wherein the wall portion includes a back wall, an upper wall, a lower wall, and a pair of side walls; the back wall closes the other end portion of the wall portion; and one end portion of the upper wall, one end portion of the lower wall, and one end portions of the side walls form the container main body opening portion, wherein a front end portion and a rear end portion of a central fixed member are provided to the wall portion of the container main body so as to be removably fixed to a central portion of the lower wall by being engaged with or disengaged with the central portion of the lower wall of the wall portion of the container main body, the central fixed member being configured to be locked and fixed to an alignment portion provided at a load port for conveying the substrate stored in the substrate storing space to a processing apparatus, and wherein a front side fixing portion and a rear side fixing portion are disposed at the central portion of the lower wall, the front side fixing portion and the rear side fixing portion are configured to removably fix the central fixed member.
2. The substrate storing container according to claim 1, wherein the central fixed member is fixed to the wall portion of the container main body by a fitting structure.
3. The substrate storing container according to claim 2, wherein the central fixed member

comprises: a member front portion having an oval shape constituting a portion of the central fixed member on one end portion side of the container main body; and a member rear portion having a rectangular shape constituting a portion of the central fixed member on the other end portion side of the container main body.

4. The substrate storing container according to claim 3, wherein the member front portion is engaged with and fixed to the container main body at one position, the member rear portion is engaged with and fixed to the container main body at two positions, and the central fixed member is fitted into the wall portion of the container main body.

5. The substrate storing container according to claim 1, wherein the central fixed member is made of a polycarbonate resin.

6. The substrate storing container according to claim 1, wherein the central fixed member is made of a cycloolefin polymer resin.
